

Package ‘ndm’

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Title Neural Disease Modeling Software

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Description Provides an R-first interface for neural disease modeling with a reticulate-managed JAX backend. The package includes backend bootstrap helpers, built-in epidemiological model specifications, TFRecord-consuming data interfaces, package-managed runtime helpers, and package-native orchestration helpers for model execution.

Depends R (>= 4.1.0)

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ndmdatasets,
rlang,
reticulate,
utils

Suggests fastmatch,
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Contents

ndm-package	2
ndm_bootstrap_sim_tfrecords	3
ndm_build_model	4
ndm_check_backend	6

ndm_create_config	8
ndm_create_real_run_config	9
ndm_create_tfrecord_parser	12
ndm_extract_temporal_latents	13
ndm_model_spec_from_structure	15
ndm_model_spec_presets	16
ndm_new_runtime_env	19
ndm_predict	19
ndm_prepare_runtime	21
ndm_print	22
ndm_save_model	22
ndm_tfrecord_fields_real	24
ndm_tf_batch_to_r	24

Index	27
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ndm-package	<i>Neural Disease Modeling Software</i>
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Description

ndm is an R-first package for neural disease modeling with a reticulate-managed JAX backend. The package currently provides backend setup, epidemic model spec import/export, TFRecord readers, package-managed runtime helpers, package-native runtime/model helpers, package-native run APIs, and artifact helpers for saving and restoring trained models.

Details

The current implementation preserves both historical model families, "DecoderOnly" and "NeuralODE", while defaulting to "DecoderOnly". The supported backbone is the transformer path only. Latent attention and Mamba are intentionally excluded.

The package treats the structured `ndm_model_spec` object as the canonical internal representation and supports `.tex` import/export for compatibility with the original epidemic specification workflow.

For executable workflows rather than dry-run previews, install the helper packages listed under [Suggests](#), initialize the backend with [ndm_initialize_backend](#), and align that backend with `NDM_SOFTWARE_CONDA_ENV` or `NDM_CONDA_ENV` when using the package-native runner helpers. When both variables are unset, the runner layer currently falls back to `jax_cpu` on macOS and `ndm_software_env` elsewhere.

Key functions

- [ndm_create_config](#)
- [ndm_build_backend](#)
- [ndm_initialize_backend](#)
- [ndm_model_spec](#)
- [ndm_model_spec_from_tex](#)
- [ndm_model_spec_to_tex](#)
- [ndm_read_tfrecord_dataset](#)

- [ndm_load_tfrecord_bundle](#)
- [ndm_prepare_runtime](#)
- [ndm_prepare_data](#)
- [ndm_build_model](#)
- [ndm_train](#)
- [ndm_predict](#)
- [ndm_loss](#)
- [ndm_fit](#)
- [ndm_create_real_run_config](#)
- [ndm_run_real](#)
- [ndm_run_sim](#)
- [ndm_run_multidisease](#)
- [ndm_save_model](#)
- [ndm_load_model](#)
- [ndm_resume_training](#)

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ndm_bootstrap_sim_tfrecords

Bootstrap canonical simulation TFRecords by BaseID

Description

This helper materializes canonical simulation TFRecords without going through the legacy ‘outer’ scheduling path. It reads the simulation grid in raw CSV or data-frame order, builds one canonical write plan row per ‘BaseID’, and optionally writes those TFRecords with a per-‘BaseID’ cross-process lock.

Usage

```
ndm_bootstrap_sim_tfrecords(  
    project_root = getwd(),  
    analysis_name = "BigSimsLatest",  
    grid = NULL,  
    grid_file = file.path("Data", "RunGrids", "SimGrids", sprintf("SimGrid_%s.csv",  
        analysis_name)),  
    base_ids = NULL,  
    tfrecord_dir = file.path("Data", "RunTFRecords", "SimTFRecords", analysis_name),  
    overwrite = FALSE,  
    dry_run = FALSE  
)
```

Arguments

project_root	Working project directory used to resolve relative grid and TFRecord paths.
analysis_name	Analysis label used to derive default grid and TFRecord paths.
grid	Optional in-memory simulation grid. When supplied, 'grid_file' must be 'NULL'.
grid_file	Optional CSV path for the simulation grid.
base_ids	Optional integer vector of canonical 'BaseID' values to materialize. When 'NULL', bootstrap all 'BaseID's in raw grid order of first appearance.
tfrecord_dir	Output directory for canonical TFRecords.
overwrite	Logical scalar controlling whether existing canonical TFRecords should be re-generated.
dry_run	Logical scalar indicating whether to return the canonical write plan without writing TFRecords.

Value

A data frame with one row per canonical 'BaseID' and columns 'BaseID', 'selected_rows', 'canonical_row', 'artifact_n_samples_train', 'train_file', 'inference_file', and 'status'.

ndm_build_model	<i>Build and train models with package-managed runtime stages</i>
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Description

These wrappers layer a small R API over package-managed Phase 1 model build and training stages.

Usage

```
ndm_build_model(
  runtime_env,
  model_type = c("DecoderOnly", "NeuralODE"),
  model_spec = NULL,
  backbone = "transformer",
  runtime_globals = list()
)

## S3 method for class 'ndm_model'
print(x, ...)

ndm_train(x, run_define = TRUE, run_loop = TRUE)

## S3 method for class 'ndm_trained_model'
print(x, ...)

ndm_fit(
  config = ndm_create_config(),
  model_spec = NULL,
  data_generator = c("sim", "real", "multidisease"),
  runtime_globals = list(),
```

```

    data_globals = list(),
    build_globals = list(),
    train_globals = list(),
    run_define = TRUE,
    run_loop = TRUE
  )

```

Arguments

<code>runtime_env</code>	Runtime environment containing loaded runtime helpers and data globals.
<code>model_type</code>	Model family to build. Either <code>"DecoderOnly"</code> or <code>"NeuralODE"</code> .
<code>model_spec</code>	Optional <code>'ndm_model_spec'</code> object used to override the model TeX specification supplied to the runtime model builder.
<code>backbone</code>	Backbone family. Phase 1 supports <code>"transformer"</code> only.
<code>runtime_globals</code>	Named list of additional globals assigned during the relevant runtime stage. <code>'ndm_build_model()'</code> uses them before building the model, while <code>'ndm_fit()'</code> uses them before sourcing the runtime.
<code>x</code>	Either an <code>'ndm_model'</code> , an <code>'ndm_trained_model'</code> , or a prepared runtime environment, depending on the function being called.
<code>...</code>	Unused; included for S3 method compatibility.
<code>run_define</code>	Logical scalar indicating whether the training definition script should be sourced.
<code>run_loop</code>	Logical scalar indicating whether the training loop script should be sourced.
<code>config</code>	An object of class <code>'ndm_config'</code> , usually created by <code>'ndm_create_config()'</code> .
<code>data_generator</code>	Which local data generator to source before calling <code>'ndm_build_model()'</code> inside <code>'ndm_fit()'</code> .
<code>data_globals</code>	Named list of globals assigned before sourcing the data generator in <code>'ndm_fit()'</code> .
<code>build_globals</code>	Named list of globals assigned before building the model in <code>'ndm_fit()'</code> .
<code>train_globals</code>	Named list of globals assigned immediately before training in <code>'ndm_fit()'</code> .

Details

These functions assume the backend has already been initialized with `'ndm_initialize_backend()'` and that `'runtime_env'` already contains the upstream runtime and data globals required by the selected workflow. `'ndm_train()'` requires `'rapply'`; it also requires `'zip'` when checkpointing is enabled and `'zoo'` for multidisease training. Simulation paths inherit the `'progress'` and `'zoo'` requirements documented on `[ndm_prepare_runtime()]`.

Value

`'ndm_build_model()'` returns an object of class `'ndm_model'`. `'ndm_train()'` and `'ndm_fit()'` return an object of class `'ndm_trained_model'`.

`'print.ndm_model()'` prints a brief summary of an `'ndm_model'` object and invisibly returns `'x'`.

`'print.ndm_trained_model()'` prints a brief summary of an `'ndm_trained_model'` object and invisibly returns `'x'`.

Examples

```
cfg <- ndm_create_config()
spec <- ndm_model_spec()
## Not run:
runtime_env <- ndm_prepare_runtime(cfg)
ndm_prepare_data(runtime_env, generator = "sim")
model <- ndm_build_model(runtime_env, model_spec = spec)
trained <- ndm_train(model)

## End(Not run)
```

ndm_check_backend	<i>Provision and initialize the Python backend</i>
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Description

These helpers manage the reticulate-backed Python environment used by ndm. Use ‘ndm_build_backend()’ to provision dependencies, ‘ndm_check_backend()’ to verify availability, and ‘ndm_initialize_backend()’ to import the active JAX stack into R.

Usage

```
ndm_check_backend(
  conda_env = "ndm_software_env",
  conda = "auto",
  modules = c("jax", "numpy", "optax", "equinox", "diffrax")
)

ndm_build_backend(
  conda_env = "ndm_software_env",
  conda = "auto",
  python_version = "3.13",
  include_tensorflow = TRUE,
  extra_packages = c("optax", "equinox", "diffrax")
)

ndm_initialize_backend(
  conda_env = "ndm_software_env",
  conda = "auto",
  conda_env_required = TRUE,
  float_type = c("32", "64"),
  import_tensorflow = TRUE
)

ndm_backend_modules()
```

Arguments

conda_env	Name of the conda environment that should contain the JAX runtime.
conda	Conda executable to use. "auto" delegates discovery to reticulate.

<code>modules</code>	Character vector of Python module names that must be importable for the backend to be considered healthy.
<code>python_version</code>	Python version requested when creating the conda environment.
<code>include_tensorflow</code>	Logical scalar indicating whether TensorFlow should be installed alongside JAX.
<code>extra_packages</code>	Additional Python packages to install with ‘pip’.
<code>conda_env_required</code>	Passed through to ‘reticulate::use_condaenv()’ as the ‘required’ flag.
<code>float_type</code>	Floating point precision used when configuring JAX. Either “32” or “64”.
<code>import_tensorflow</code>	Logical scalar indicating whether TensorFlow should be imported when it is available in the selected environment.

Details

When you want backend helpers and the package-native ‘`ndm_run_*`’ runners to target the same conda environment, set ‘`NDM_SOFTWARE_CONDA_ENV`’ (or ‘`NDM_CONDA_ENV`’) once and pass that value into these helpers. The runner layer currently falls back to ‘`jax_cpu`’ on macOS and ‘`ndm_software_env`’ elsewhere when neither environment variable is set.

Value

‘`ndm_check_backend()`’ returns ‘`TRUE`’ invisibly when the requested environment and modules are available. It returns ‘`NULL`’ after printing a diagnostic message when the backend is not ready.

‘`ndm_build_backend()`’ invisibly returns the conda environment name after attempting to provision the requested packages.

‘`ndm_initialize_backend()`’ returns an object of class ‘`ndm_backend`’. The returned list contains imported Python modules, the configured float type, and a small compatibility shim used by the legacy runtime.

‘`ndm_backend_modules()`’ returns the currently initialized ‘`ndm_backend`’ object.

Examples

```
ndm_check_backend()

## Not run:
ndm_build_backend()

## End(Not run)

## Not run:
backend <- ndm_initialize_backend()
backend$default_backend

## End(Not run)

## Not run:
ndm_initialize_backend()
ndm_backend_modules()

## End(Not run)
```

ndm_create_config *Create runtime configuration objects*

Description

Configuration objects collect the runtime options that are threaded through the higher-level orchestration helpers such as ‘ndm_prepare_runtime()’ and ‘ndm_fit()’.

Usage

```
ndm_create_config(
  model_type = c("DecoderOnly", "NeuralODE"),
  backbone = "transformer",
  float_type = c("32", "64"),
  force_to_gpu = TRUE,
  resave_tfrecords = FALSE,
  gpu_mem_frac = NULL,
  neuralode_local_latent_dim = NULL,
  neuralode_global_latent_dim = NULL,
  neuralode_init_state_logit_offset = NULL,
  neuralode_init_state_logit_scale_max = Inf,
  neuralode_optim_solver = c("tsit5", "dopri8"),
  neuralode_optim_dt0 = 0.001,
  neuralode_optim_controller = c("pid", "constant"),
  neuralode_optim_rtol = 1e-05,
  neuralode_optim_atol = 1e-07,
  ...
)

## S3 method for class 'ndm_config'
print(x, ...)
```

Arguments

model_type	Model family metadata. Use “DecoderOnly” or “NeuralODE”.
backbone	Backbone family. Phase 1 supports “transformer” only.
float_type	Floating point precision used when initializing the backend. Use “32” or “64”.
force_to_gpu	Logical scalar indicating whether the runtime should try to place arrays on GPU when available.
resave_tfrecords	Logical scalar preserved for compatibility with existing run workflows. Multidisease workflows reject ‘TRUE’ because the legacy TFRecord-regeneration path has been retired.
gpu_mem_frac	Optional GPU memory fraction forwarded into the runtime bootstrap code.
neuralode_local_latent_dim	Optional local NeuralODE latent width. When ‘NULL’, runtime code resolves this to ‘ModelDims’.

neuralode_global_latent_dim	Optional global NeuralODE latent width. When 'NULL', runtime code resolves this to 'ModelDims'.
neuralode_init_state_logit_offset	Optional init-state logit offset vector for the NeuralODE state simplex initializer.
neuralode_init_state_logit_scale_max	Scalar cap applied to the NeuralODE init-state logit scale.
neuralode_optim_solver	NeuralODE optimization solver name. Use "tsit5" or "dopri8".
neuralode_optim_dt0	Initial step size used by the NeuralODE optimization solve.
neuralode_optim_controller	NeuralODE optimization stepsize controller. Use "pid" or "constant".
neuralode_optim_rtol	Relative tolerance used when 'neuralode_optim_controller = "pid"'.
neuralode_optim_atol	Absolute tolerance used when 'neuralode_optim_controller = "pid"'.
...	Unused; included for S3 method compatibility.
x	An 'ndm_config' object.

Value

'ndm_create_config()' returns an object of class 'ndm_config'.

'print.ndm_config()' prints a brief summary of an 'ndm_config' object and invisibly returns 'x'.

Examples

```
cfg <- ndm_create_config()
print(cfg)
```

ndm_create_real_run_config

Package-native run configurations and workflow runners

Description

Package-native configuration constructors and workflow runners for the self-contained real, simulation, and multidisease execution paths.

Usage

```
ndm_create_real_run_config(
  project_root = getwd(),
  analysis_name = "RealApril15",
  grid = NULL,
  grid_file = file.path("Data", "RunGrids", "RealGrids", sprintf("RealGrid_%s.csv",
    analysis_name)),
  outer = 3L,
```

```

    model_type = NULL,
    respect_grid_model_type = FALSE,
    resave_tfrerecords = FALSE,
    tfrecord_dir = file.path("Data", "RunTFRecords", "RealTFRecords",
        analysis_name),
    raw_data_dir = file.path("Data", "MainData"),
    outcome_metric = "inc_death",
    data_subset = "high_income",
    dry_run = FALSE
)

ndm_create_sim_run_config(
    project_root = getwd(),
    analysis_name = "BigSimsLatest",
    grid = NULL,
    grid_file = file.path("Data", "RunGrids", "SimGrids", sprintf("SimGrid_%s.csv",
        analysis_name)),
    outer = 1L,
    model_type = NULL,
    respect_grid_model_type = FALSE,
    resave_tfrerecords = TRUE,
    tfrecord_dir = file.path("Data", "RunTFRecords", "SimTFRecords",
        analysis_name),
    dry_run = FALSE
)

ndm_create_multidisease_run_config(
    project_root = getwd(),
    analysis_name = "RealLatest",
    grid = NULL,
    grid_file = file.path("Data", "RunGrids", "RealGrids", sprintf("RealGrid_%s.csv",
        analysis_name)),
    outer = 1L,
    model_type = NULL,
    respect_grid_model_type = FALSE,
    resave_tfrerecords = FALSE,
    tfrecord_dir = file.path("Data", "RunTFRecords", "RealTFRecords",
        analysis_name),
    outcome_metric = "CountValue",
    data_subset = "all",
    disease_names = c("Covid", "Flu"),
    data_format = "IHME",
    dry_run = FALSE
)

ndm_run_real(config = ndm_create_real_run_config())

ndm_run_sim(config = ndm_create_sim_run_config())

ndm_run_multidisease(config = ndm_create_multidisease_run_config())

```

Arguments

<code>project_root</code>	Working project directory used for grids, TFRecords, and outputs.
<code>analysis_name</code>	Analysis label threaded through output paths.
<code>grid</code>	Optional in-memory grid data frame. When supplied, <code>grid_file</code> must be NULL.
<code>grid_file</code>	Optional CSV path for the run grid.
<code>outer</code>	Integer vector of outer-iteration rows to execute.
<code>model_type</code>	Optional model family override.
<code>respect_grid_model_type</code>	Logical scalar indicating whether grid rows may override the requested model type.
<code>resave_tfrecords</code>	Logical scalar controlling TFRecord regeneration. Supported for real and simulation workflows. Multidisease workflows reject TRUE because the legacy regeneration path has been retired.
<code>tfrecord_dir</code>	Optional TFRecord output directory.
<code>raw_data_dir</code>	Real-data input directory.
<code>outcome_metric</code>	Outcome metric name for real-data or multidisease runs.
<code>data_subset</code>	Optional real-data or multidisease subset name.
<code>disease_names</code>	Character vector of multidisease names.
<code>data_format</code>	Multidisease data format.
<code>dry_run</code>	Logical scalar indicating whether the run should stop after resolving paths and grid rows.
<code>config</code>	A package-native run configuration created by the matching <code>ndm_create_*_run_config()</code> helper.

Details

`dry_run = TRUE` resolves paths and selected rows without executing the underlying workflow. Dry runs can use lightweight in-memory grids such as the BaseID / ModelType preview shown in `README.md`.

Non-dry execution expects Analysis2-compatible grids. The tested simulation grids in this package are produced by `ndmdatasets::ndm_sim_build_grid()` and then augmented with fields such as `paddingMethod`, `lookahead`, `n_time_steps`, `n_inference_batches`, `scaling_batches`, and either `model_spec_name` or `model_tex_loc`. The tested real and multidisease grids include BaseID, ContextLength, evaluationTime, initialTransform, initialNormType, paddingMethod, OSSType, dataInputs, ModelType, ModelDepth, ModelDims, nSamplesTrain, nObsInference, floatType, and either `model_spec_name` or `model_tex_loc`.

Non-dry runner workflows also require the full execution stack: the `ndmdatasets` package, the helper packages used by the runtime, and a backend environment aligned with `NDM_SOFTWARE_CONDA_ENV`, `NDM_CONDA_ENV`, or the platform-specific fallback used by the runner layer.

Value

The `ndm_create_*_run_config()` helpers return classed lists describing the requested workflow. `ndm_run_real()` and `ndm_run_sim()` return a dry-run preview when `config$dry_run` is TRUE, a list containing `written_base_ids`, `tfrecord_dir`, and `write_plan` when `config$resave_tfrecords` is TRUE, and otherwise the underlying workflow result. `ndm_run_multidisease()` returns a dry-run preview when `config$dry_run` is TRUE and otherwise the underlying workflow result.

Examples

```
## Not run:
preview_grid <- data.frame(
  BaseID = c(1L, 2L),
  ModelType = c("DecoderOnly", "NeuralODE"),
  stringsAsFactors = FALSE
)

cfg <- ndm_create_sim_run_config(
  project_root = tempdir(),
  grid = preview_grid,
  outer = 1:2,
  dry_run = TRUE
)

ndm_run_sim(cfg)

## End(Not run)
```

ndm_create_tfrecord_parser

Create and read TensorFlow TFRecord datasets

Description

These wrappers construct parsers for the ndm TFRecord schema and use them to build TensorFlow datasets or eagerly collect batches into R.

Usage

```
ndm_create_tfrecord_parser(field_names, dtype_map = NULL, tensorflow = NULL)

ndm_read_tfrecord_dataset(
  file,
  batch_size,
  field_names,
  dtype_map = NULL,
  max_examples = NULL,
  shuffle = FALSE,
  buffer_scaler = 10L,
  tensorflow = NULL
)

ndm_collect_tfrecord_batches(
  file,
  batch_size,
  field_names,
  dtype_map = NULL,
  max_batches = NULL,
  max_examples = NULL,
  shuffle = FALSE,
  buffer_scaler = 10L,
```

```
    tensorflow = NULL
  )
```

Arguments

field_names	Character vector of TFRecord field names that should be parsed from each example.
dtype_map	Optional named vector or list mapping 'field_names' to TensorFlow dtype names or TensorFlow dtype objects.
tensorflow	Optional TensorFlow module. When omitted, ndm uses the active backend's TensorFlow module when available and otherwise imports TensorFlow from the active reticulate environment at call time.
file	Path to the '.tfrecord' file to read.
batch_size	Batch size used when constructing the TensorFlow dataset.
max_examples	Optional maximum number of examples to consume from the file before batching.
shuffle	Logical scalar indicating whether the dataset should be shuffled.
buffer_scaler	Integer multiplier used to derive the TensorFlow shuffle buffer size from 'batch_size'.
max_batches	Optional maximum number of batches to collect when using 'ndm_collect_tfrecord_batches()'.

Details

TFRecord parsing requires TensorFlow to be importable from the active reticulate environment. In practice this usually means provisioning the backend with 'ndm_build_backend(include_tensorflow = TRUE)' and then calling 'ndm_initialize_backend(import_tensorflow = TRUE)' for the same conda environment before reading datasets.

Value

'ndm_create_tfrecord_parser()' returns a parser function compatible with 'tf\$data\$Dataset\$map()'. 'ndm_read_tfrecord_dataset()' returns a TensorFlow dataset object. 'ndm_collect_tfrecord_batches()' returns a list of parsed batches.

Examples

```
fields <- ndm_tfrecord_fields_real()
parser <- ndm_create_tfrecord_parser(fields, ndm_tfrecord_dtype_map("real"))
is.function(parser)
```

ndm_extract_temporal_latents

Extract temporal latent states for interpretability analyses

Description

'ndm_extract_temporal_latents()' runs the existing prediction path and returns aligned temporal representations for decoder-only and NeuralODE models. 'ndm_latent_matrix()' flattens one extracted representation to a two-dimensional matrix suitable for external analyses such as CCA.

Usage

```

ndm_extract_temporal_latents(
  x,
  batch = NULL,
  seed = 1L,
  inference = TRUE,
  update_state = FALSE,
  as_r = TRUE
)

ndm_latent_matrix(latents, source = "decoder_head_input", features = NULL)

```

Arguments

x	Either an ‘ndm_model’, an ‘ndm_trained_model’, or a prepared runtime environment that already contains the required runtime objects.
batch	Optional prediction batch passed through to [ndm_predict()].
seed	Integer seed used to derive the per-example JAX RNG keys.
inference	Logical scalar indicating whether the inference prediction path or training prediction path should be used.
update_state	Logical scalar indicating whether the runtime ‘state’ object should be updated with the returned state.
as_r	Logical scalar. When ‘TRUE’, JAX arrays are copied to R arrays.
latents	An object returned by ‘ndm_extract_temporal_latents()’.
source	Representation to flatten. Supported values include “decoder_head_input”, “ode_local_hidden”, “ode_global_hidden”, “ode_local_features”, “ode_global_features”, and “ode_states”.
features	Optional feature names or integer feature positions.

Value

‘ndm_extract_temporal_latents()’ returns an ‘ndm_temporal_latents’ list. ‘ndm_latent_matrix()’ returns a numeric matrix with rows ordered by batch within time.

Examples

```

## Not run:
latents <- ndm_extract_temporal_latents(trained, batch = batch_1_cal)
decoder_x <- ndm_latent_matrix(latents, "decoder_head_input")
ode_x <- ndm_latent_matrix(latents, "ode_local_hidden")

## End(Not run)

```

ndm_model_spec_from_structure

Inspect and create epidemic model specifications from structured declarations

Description

`'ndm_model_spec_from_structure()'` builds an `'ndm_model_spec'` from a structured declaration. The declaration may either describe the full model directly or identify a supported built-in family plus optional `'family_args'` that expand parameterized structures into explicit executable equations.

Usage

```
ndm_model_spec_from_structure(
  structure,
  model_type = c("DecoderOnly", "NeuralODE"),
  family_args = NULL
)
```

Arguments

<code>structure</code>	A structured model declaration.
<code>model_type</code>	Model family metadata attached to the returned specification. Either <code>"DecoderOnly"</code> or <code>"NeuralODE"</code> .
<code>family_args</code>	Optional named list used to expand parameterized family declarations. For structured TB families, <code>'observation_target'</code> chooses <code>"incidence"</code> (default flow into <code>'i_1'</code>) or <code>"cumulative"</code> (<code>'i_1'</code>). TB families use <code>'dynamics_variant = "balanced_incidence"</code> by default, preserving the original active-progression flow while adding normalized at-risk renewal (<code>'N = 1'</code>), active-disease exit back to susceptibility, latent mortality, and incidence-flow observations. Set <code>'dynamics_variant = "progression"</code> for legacy one-way depletion dynamics. Balanced TB specs require incidence observations.

Details

For a maintainer-oriented guide to the required fields, parameter-spec shape, TeX contract, and extension workflow, see `vignette("ode-structures", package = "ndm")`.

Value

An object of class `'ndm_model_spec'`.

Examples

```
family_spec <- ndm_model_spec_from_structure(
  list(
    family = "tb_b",
    family_args = list(n = 4L)
  ),
  model_type = "NeuralODE"
)
family_spec$state_terms
```

```

toy_spec <- ndm_model_spec_from_structure(
  list(
    preset = "toy_slqi",
    description = "Toy four-state latent structure with a holding compartment.",
    states = c("s_l", "l_l", "q_l", "i_l"),
    parameters = list(
      lambda = list(
        transformation = "InvSoftPlus",
        prior_mean = 0.08,
        prior_sd = 0.25
      ),
      c = list(
        transformation = "InvSoftPlus",
        prior_mean = 0.05,
        prior_sd = 0.25
      ),
      h = list(
        transformation = "InvSoftPlus",
        prior_mean = 0.03,
        prior_sd = 0.25
      )
    ),
    equations = c(
      s_l = "- lambda * s_l",
      l_l = "lambda * s_l - (c + h) * l_l",
      q_l = "h * l_l",
      i_l = "c * l_l"
    ),
    observations = "i_l"
  ),
  model_type = "NeuralODE"
)
toy_spec$parameter_terms

```

ndm_model_spec_presets

Inspect and create epidemic model specifications

Description

The package ships a small registry of built-in compartmental model specifications and also supports importing custom ‘.tex’ model definitions from disk. The structured ‘ndm_model_spec’ object is the canonical representation used by the runtime wrappers.

Usage

```

ndm_model_spec_presets()

ndm_model_spec(
  preset = NULL,
  compartments = c("seir", "seirs"),
  dynamic_beta = NULL,

```



```

dynamic_global = FALSE,
multi_outcome = FALSE,
model_type = c("DecoderOnly", "NeuralODE"),
time_varying_terms = NULL,
endogenous_terms = NULL,
tex_path = NULL,
family_args = NULL
)

ndm_model_spec_from_tex(path, model_type = c("DecoderOnly", "NeuralODE"))

ndm_model_spec_to_tex(spec, path = NULL)

## S3 method for class 'ndm_model_spec'
print(x, ...)
```

Arguments

preset	Optional preset name from <code>'ndm_model_spec_presets()'</code> . When omitted, the remaining arguments are used to infer a built-in preset.
compartments	Compartment family used when inferring a built-in preset.
dynamic_beta	Logical scalar controlling whether the local transmission rate is time-varying for built-in presets.
dynamic_global	Logical scalar controlling whether the global rates are time-varying for built-in presets.
multi_outcome	Logical scalar indicating whether the observation model should include multiple outcomes.
model_type	Model family metadata attached to the returned specification. Either <code>"DecoderOnly"</code> or <code>"NeuralODE"</code> .
time_varying_terms	Optional character vector overriding the default time-varying term metadata stored on the returned spec.
endogenous_terms	Optional character vector overriding the default endogenous term metadata stored on the returned spec.
tex_path	Optional path to a custom <code>'tex'</code> file. When supplied, <code>'ndm_model_spec()'</code> delegates to <code>'ndm_model_spec_from_tex()'</code> .
family_args	Optional named list used to expand parameterized model families such as chained latent-stage specifications. For structured TB presets, <code>'n'</code> controls supported chain lengths and <code>'observation_target'</code> chooses <code>"incidence"</code> (default flow into <code>'i_l'</code>) or <code>"cumulative"</code> (<code>'i_l'</code>). TB presets use <code>'dynamics_variant = "balanced_incidence"</code> by default to keep the original progression topology while adding demographic renewal, active-disease exit back to susceptibility, and incidence-flow observations. Set <code>'dynamics_variant = "progression"</code> for legacy one-way depletion dynamics. Balanced TB specs require incidence observations.
path	Path to a <code>'tex'</code> model specification file on disk.
spec	An <code>'ndm_model_spec'</code> object.
x	An <code>'ndm_model_spec'</code> object.
...	Unused; included for S3 method compatibility.

Details

For a maintainer-oriented walkthrough of preset selection, structured families, TeX round-tripping, and safe modification workflows, see `vignette("ode-structures", package = "ndm")`.

Value

`'ndm_model_spec_presets()'` returns a data frame describing the built-in presets bundled with the package.

`'ndm_model_spec()'` returns an object of class `'ndm_model_spec'`. Built-in presets include meta-data such as the packaged source path and the normalized TeX contents.

`'ndm_model_spec_from_tex()'` returns an `'ndm_model_spec'` object whose `'source_path'` points at the supplied file. When the basename matches a built-in preset, the preset metadata is reused.

`'ndm_model_spec_to_tex()'` returns the TeX text as a single character string when `'path'` is `'NULL'`. When `'path'` is supplied, it writes the specification to disk and invisibly returns the normalized output path.

`'print.ndm_model_spec()'` prints a brief summary of an `'ndm_model_spec'` object and invisibly returns `'x'`.

Examples

```
ndm_model_spec_presets()

spec <- ndm_model_spec()
print(spec)

tb_spec <- ndm_model_spec(
  preset = "tb_b",
  model_type = "NeuralODE",
  family_args = list(n = 4L)
)
tb_spec$family_args

spec <- ndm_model_spec()
tex_path <- tempfile(fileext = ".tex")
ndm_model_spec_to_tex(spec, tex_path)
ndm_model_spec_from_tex(tex_path)

tb_spec <- ndm_model_spec(
  preset = "tb_k",
  model_type = "NeuralODE"
)
tb_tex_path <- tempfile(fileext = ".tex")
ndm_model_spec_to_tex(tb_spec, tb_tex_path)
ndm_model_spec_from_tex(tb_tex_path, model_type = "NeuralODE")

spec <- ndm_model_spec()
tex <- ndm_model_spec_to_tex(spec)
nchar(tex) > 0
```

ndm_new_runtime_env *Create and populate runtime environments*

Description

These helpers manage isolated environments used to load package-managed runtime code and seed it with R values.

Usage

```
ndm_new_runtime_env(parent = globalenv())

ndm_set_runtime_globals(env, values, overwrite = TRUE)
```

Arguments

parent	Parent environment for a new runtime environment. Defaults to ‘globalenv()’ so package-managed runtime code can resolve attached-package functions consistently.
env	Runtime environment that should receive new bindings.
values	Named list of values to assign into ‘env’.
overwrite	Logical scalar indicating whether existing bindings in ‘env’ should be overwritten.

Value

‘ndm_new_runtime_env()’ returns an environment of class ‘ndm_runtime_env’. ‘ndm_set_runtime_globals()’ invisibly returns ‘env’.

Examples

```
env <- ndm_new_runtime_env()
ndm_set_runtime_globals(env, list(example_value = 1L))
env$example_value
```

ndm_predict *Generate predictions or evaluate the training loss*

Description

These helpers call the runtime prediction and loss functions stored in a prepared runtime environment.

Usage

```
ndm_predict(x, batch = NULL, inference = TRUE, seed = 1L, update_state = TRUE)

ndm_loss(
  x,
  batch = NULL,
  y = NULL,
  y_mask = NULL,
  iteration = 1L,
  seed = 1L,
  update_state = TRUE
)
```

Arguments

<code>x</code>	Either an ‘ndm_model’, an ‘ndm_trained_model’, or a prepared runtime environment that already contains the required runtime objects.
<code>batch</code>	Optional batch object. Supply either a named TFRecord-style list or the four-element packaged list returned by ‘ndm_batch_to_model_inputs()’. When ‘NULL’, the runtime must already contain ‘batch_1_cal’.
<code>inference</code>	Logical scalar indicating whether ‘ndm_predict()’ should use the inference prediction path or the training prediction path.
<code>seed</code>	Integer seed used to derive the per-example JAX RNG keys.
<code>update_state</code>	Logical scalar indicating whether the runtime ‘state’ object should be updated with the returned state.
<code>y</code>	Optional observed outcome tensor passed to ‘ndm_loss()’. When omitted, ‘YTrue_out’ is inferred from ‘batch’.
<code>y_mask</code>	Optional outcome mask tensor passed to ‘ndm_loss()’. When omitted, ‘YTrue_out_mask’ is inferred from ‘batch’.
<code>iteration</code>	Training iteration number forwarded to the runtime loss function.

Value

‘ndm_predict()’ returns the prediction object produced by the runtime. ‘ndm_loss()’ returns the loss object produced by the runtime.

Examples

```
## Not run:
preds <- ndm_predict(model, batch = batch_1_cal)
loss <- ndm_loss(model, batch = batch_1_cal)

## End(Not run)
```

ndm_prepare_runtime *Prepare a local runtime for execution*

Description

These wrappers load package-owned runtime components into an isolated environment and then source the selected data generator.

Usage

```
ndm_prepare_runtime(
  config = ndm_create_config(),
  runtime_env = ndm_new_runtime_env(),
  runtime_globals = list()
)

ndm_prepare_data(
  runtime_env,
  generator = c("sim", "real", "multidisease"),
  runtime_globals = list()
)
```

Arguments

config	An object of class 'ndm_config', usually created by 'ndm_create_config()'.
runtime_env	Runtime environment that should receive the sourced objects.
runtime_globals	Named list of additional bindings to assign before the runtime code is sourced.
generator	Which data generator to source into 'runtime_env'.

Details

Before building, training, predicting, or restoring artifacts from a runtime environment, initialize the backend for the target conda environment with 'ndm_initialize_backend()'. 'ndm_prepare_runtime()' requires the 'fastmatch' package. 'ndm_prepare_data(generator = "sim")' additionally requires 'progress' and 'zoo'.

Value

'ndm_prepare_runtime()' invisibly returns 'runtime_env' after loading the package-owned helper and backend code.

'ndm_prepare_data()' invisibly returns 'runtime_env' after sourcing the requested data generator.

Examples

```
env <- ndm_new_runtime_env()
cfg <- ndm_create_config()
## Not run:
ndm_prepare_runtime(cfg, env)

## End(Not run)
```

```
env <- ndm_new_runtime_env()
ndm_set_runtime_globals(env, list(example_value = 1L))
```

ndm_print	<i>Print a timestamped diagnostic message</i>
-----------	---

Description

‘ndm_print()’ is a small logging helper used throughout the package-facing orchestration code.

Usage

```
ndm_print(text, quiet = FALSE)
```

Arguments

text	Text to print.
quiet	Logical scalar indicating whether output should be suppressed.

Value

The input ‘text’, invisibly.

Examples

```
ndm_print("starting")
```

ndm_save_model	<i>Save and restore model artifacts</i>
----------------	---

Description

These helpers persist the public ‘ndm_model’ / ‘ndm_trained_model’ objects to a versioned artifact directory, restore them into a fresh runtime environment, and resume training from the restored optimizer state.

Usage

```
ndm_save_model(x, path, bundle = NULL, overwrite = FALSE)
```

```
ndm_load_model(
  path,
  bundle = NULL,
  calibration_source = c("inference", "train")
)
```

```
ndm_resume_training(
```

```

    path,
    bundle = NULL,
    calibration_source = c("train", "inference"),
    train_globals = list()
)

```

Arguments

<code>x</code>	An object of class <code>'ndm_model'</code> or <code>'ndm_trained_model'</code> .
<code>path</code>	Artifact directory path.
<code>bundle</code>	Optional <code>'ndm_tfrecord_bundle'</code> used to record TFRecord references for later resume workflows.
<code>overwrite</code>	Logical scalar indicating whether an existing artifact directory should be replaced.
<code>calibration_source</code>	Which dataset should supply the preview batch used to seed runtime globals such as <code>'batch_1_cal'</code> when <code>'bundle'</code> is attached.
<code>train_globals</code>	Optional named list of runtime globals applied immediately before <code>'ndm_resume_training()'</code> continues the training loop.

Details

These helpers assume `'ndm_initialize_backend()'` has already run for the active conda environment. `'ndm_save_model()'` only works for runtime-backed `'ndm_model'` / `'ndm_trained_model'` objects whose environments still contain the built model state, including `'ModelList'` and `'state'`. Exact resume also requires saved optimizer state. When `'bundle = NULL'` in `'ndm_resume_training()'`, the artifact metadata must already contain a recorded TFRecord bundle reference. `'ndm_load_model()'` and training-restore paths require the `'rrapply'` package.

Value

`'ndm_save_model()'` returns the normalized artifact directory path, invisibly. `'ndm_load_model()'` returns an `'ndm_model'` or `'ndm_trained_model'` matching the saved artifact. `'ndm_resume_training()'` returns an `'ndm_trained_model'`.

Examples

```

## Not run:
# After ndm_initialize_backend() and ndm_train() / ndm_fit():
artifact_dir <- ndm_save_model(trained_model, tempfile("ndm-artifact-"))
restored <- ndm_load_model(artifact_dir, bundle = tf_bundle)
resumed <- ndm_resume_training(artifact_dir, bundle = tf_bundle)

## End(Not run)

```

ndm_tfrecord_fields_real

Inspect TFRecord field contracts

Description

These helpers describe the field names and default TensorFlow dtypes expected by the packaged TFRecord readers.

Usage

```
ndm_tfrecord_fields_real()
```

```
ndm_tfrecord_fields_sim()
```

```
ndm_tfrecord_dtype_map(kind = c("real", "sim"))
```

Arguments

kind	TFRecord schema to inspect. Use "real" for observed-data records and "sim" for simulation records.
------	--

Value

'ndm_tfrecord_fields_real()' and 'ndm_tfrecord_fields_sim()' return character vectors of field names. 'ndm_tfrecord_dtype_map()' returns a named character vector mapping each field to its default TensorFlow dtype.

Examples

```
ndm_tfrecord_fields_real()
ndm_tfrecord_dtype_map("sim")
```

ndm_tf_batch_to_r

Convert and bundle TFRecord batches

Description

These helpers convert TensorFlow batches to native R or JAX objects, package them into the shape expected by the package-managed runtime, and attach dataset handles to a runtime environment.

Usage

```
ndm_tf_batch_to_r(batch)
```

```
ndm_tf_batch_to_jax(batch, backend = NULL)
```

```
ndm_batch_to_model_inputs(batch)
```



```

ndm_load_tfrecord_bundle(
    train_file,
    inference_file = NULL,
    batch_size,
    kind = c("real", "sim"),
    dtype_map = NULL,
    tensorflow = NULL,
    max_train_examples = NULL,
    max_inference_examples = NULL,
    shuffle_train = TRUE,
    shuffle_inference = FALSE
)

ndm_attach_tfrecord_bundle(
    env,
    bundle,
    backend = NULL,
    calibration_source = c("inference", "train")
)

```

Arguments

batch	A parsed TFRecord batch or a nested list containing TensorFlow tensors.
backend	Optional 'ndm_backend' object. When omitted, 'ndm_tf_batch_to_jax()' and 'ndm_attach_tfrecord_bundle()' use the active backend from 'ndm_backend_modules()' when available.
train_file	Path to the training TFRecord file.
inference_file	Optional path to an inference TFRecord file.
batch_size	Batch size used when constructing the training and inference datasets.
kind	TFRecord schema to use when reading 'train_file' and 'inference_file'.
dtype_map	Optional named dtype mapping passed through to the TFRecord readers.
tensorflow	Optional TensorFlow module. When omitted, ndm uses the active backend's TensorFlow module when available and otherwise imports TensorFlow from the active reticulate environment at call time.
max_train_examples	Optional cap on the number of training examples to read.
max_inference_examples	Optional cap on the number of inference examples to read.
shuffle_train	Logical scalar indicating whether the training dataset should be shuffled.
shuffle_inference	Logical scalar indicating whether the inference dataset should be shuffled.
env	Runtime environment that should receive dataset handles and the preview calibration batch.
bundle	An object of class 'ndm_tfrecord_bundle' returned by 'ndm_load_tfrecord_bundle()'.
calibration_source	Which dataset should supply the preview batch used to seed runtime globals such as 'batch_l_cal'.

Value

`'ndm_tf_batch_to_r()'` recursively converts a TensorFlow batch to R arrays and lists. `'ndm_tf_batch_to_jax()'` recursively converts tensors into JAX arrays. `'ndm_batch_to_model_inputs()'` returns the packaged four-element list expected by the package-managed model stages. `'ndm_load_tfrecord_bundle()'` returns an `'ndm_tfrecord_bundle'` object. `'ndm_attach_tfrecord_bundle()'` invisibly returns `'env'`.

Examples

```
batch <- list(
  XPred = array(0, c(2, 3, 4)),
  XPred_mask = array(TRUE, c(2, 3, 4)),
  Context = array(0, c(2, 1, 4)),
  Context_mask = array(TRUE, c(2, 1, 4)),
  location_id_numeric = matrix(1L, nrow = 2),
  time_id_numeric = matrix(1L, nrow = 2)
)
packaged <- ndm_batch_to_model_inputs(batch)
length(packaged)
```

Index

ndm (ndm-package), 2
ndm-package, 2
ndm_attach_tfrecord_bundle
 (ndm_tf_batch_to_r), 24
ndm_backend_modules
 (ndm_check_backend), 6
ndm_batch_to_model_inputs
 (ndm_tf_batch_to_r), 24
ndm_bootstrap_sim_tfrecords, 3
ndm_build_backend, 2
ndm_build_backend (ndm_check_backend), 6
ndm_build_model, 3, 4
ndm_check_backend, 6
ndm_collect_tfrecord_batches
 (ndm_create_tfrecord_parser),
 12
ndm_create_config, 2, 8
ndm_create_multidisease_run_config
 (ndm_create_real_run_config), 9
ndm_create_real_run_config, 3, 9
ndm_create_sim_run_config
 (ndm_create_real_run_config), 9
ndm_create_tfrecord_parser, 12
ndm_extract_temporal_latents, 13
ndm_fit, 3
ndm_fit (ndm_build_model), 4
ndm_initialize_backend, 2
ndm_initialize_backend
 (ndm_check_backend), 6
ndm_latent_matrix
 (ndm_extract_temporal_latents),
 13
ndm_load_model, 3
ndm_load_model (ndm_save_model), 22
ndm_load_tfrecord_bundle, 3
ndm_load_tfrecord_bundle
 (ndm_tf_batch_to_r), 24
ndm_loss, 3
ndm_loss (ndm_predict), 19
ndm_model_spec, 2
ndm_model_spec
 (ndm_model_spec_presets), 16
ndm_model_spec_from_structure, 15
ndm_model_spec_from_tex, 2
ndm_model_spec_from_tex
 (ndm_model_spec_presets), 16
ndm_model_spec_presets, 16
ndm_model_spec_to_tex, 2
ndm_model_spec_to_tex
 (ndm_model_spec_presets), 16
ndm_new_runtime_env, 19
ndm_predict, 3, 19
ndm_prepare_data, 3
ndm_prepare_data (ndm_prepare_runtime),
 21
ndm_prepare_runtime, 3, 21
ndm_print, 22
ndm_read_tfrecord_dataset, 2
ndm_read_tfrecord_dataset
 (ndm_create_tfrecord_parser),
 12
ndm_resume_training, 3
ndm_resume_training (ndm_save_model), 22
ndm_run_multidisease, 3
ndm_run_multidisease
 (ndm_create_real_run_config), 9
ndm_run_real, 3
ndm_run_real
 (ndm_create_real_run_config), 9
ndm_run_sim, 3
ndm_run_sim
 (ndm_create_real_run_config), 9
ndm_save_model, 3, 22
ndm_set_runtime_globals
 (ndm_new_runtime_env), 19
ndm_tf_batch_to_jax
 (ndm_tf_batch_to_r), 24
ndm_tf_batch_to_r, 24
ndm_tfrecord_dtype_map
 (ndm_tfrecord_fields_real), 24
ndm_tfrecord_fields_real, 24
ndm_tfrecord_fields_sim
 (ndm_tfrecord_fields_real), 24
ndm_train, 3
ndm_train (ndm_build_model), 4
print.ndm_config (ndm_create_config), 8

```
print.ndm_model(ndm_build_model), 4  
print.ndm_model_spec  
    (ndm_model_spec_presets), 16  
print.ndm_trained_model  
    (ndm_build_model), 4
```